INTERNATIONAL MONETARY FUND

GRENADA

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REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY —DEBT SUSTAINABILIT ANALYSIS

Approved By
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Prepared by the staffs of the International Monetary Fund and the International Development Association.

Risk of external debt distress:	In debt distress[1]
Overall risk of debt distress	In debt distress
Granularity in the risk rating	Sustainable
Application of judgment	No.
Macroeconomic projections	Compared to the previous DSA, there is a temporary jump in 2020 public debt ratios (to 68.7 vs. 53.7 percent of GDP) driven by lower growth (-9.2 vs. 2.7 percent), larger fiscal deficit (2.9 vs. a surplus of 4.6 percent of GDP) and a larger current account deficit (27.4 vs. 10.2 percent of GDP).
Financing strategy	The government's medium-term debt strategy prioritizes concessional external financing. The new near-term needs of COVID-19 are being met through a combination of budget reallocations, deposits and financing from development partners.
Realism tools flagged	None
Mechanical risk rating under the external DSA	High
Mechanical risk rating under the public DSA	High

[1]Grenada's debt definition in the current DSA is unchanged from the previous DSA of 2019. Grenada's Composite Indicator score is 3.03 and its debt-carrying capacity is moderate.

Grenada's external public debt is classified as "in debt distress" due to remaining unresolved arrears to official bilateral creditors of some 1.8 percent of GDP, but its debt remains sustainable, a conclusion that is unchanged from the previous DSA of 2019. The risk rating for external debt was "moderate" in the May 2019 DSA and is now "high" due to threshold breaches of the debt-services-to-exports ratio for 2020-2023, essentially due to the COVID-19 shock. The large COVID-19 related contraction in output, decline in tax revenues, and increase in health and social expenditures would cause an uptick in the debt ratio to 68¾ percent of GDP in 2020, but the subsequent economic recovery should help reverse this rise. The underlying medium-term dynamics, being anchored by the Fiscal Responsibility Law (FRL), follow a downward path. The primary fiscal surplus is projected to increase above the FRL's 3.5 percent of GDP floor after 2020 as the economy normalizes, anchoring debt sustainability. Medium-term financing needs are moderate and are expected to be covered by external borrowing with a substantial concessional component. Risks to these debt dynamics include the possibility of a more prolonged impact of COVID-19, possible delays in the return to the FRL's core parameters, natural disasters, and a one-off increase in debt if Grenlec-related payment obligations are met by debt issuance.

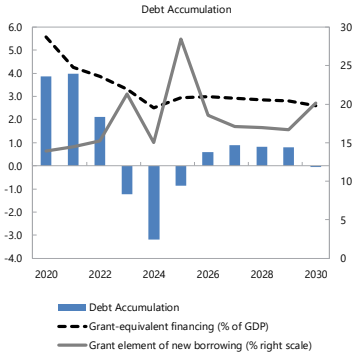
Fully regularizing external arrears would help tangibly improve the country's DSA rating. Arrears of 1.8 percent of GDP remain with three official bilateral creditors: Algeria, Libya, and Trinidad and Tobago. In mid-2019, Grenada signed the protocols on the consolidation of its debt to Algeria and reported submitting those documents to Algeria's UN mission. Grenada continued making payments on outstanding fees owed to international organizations as per schedule agreed in 2017, with some EC$13.3 million in such payments made during 2019. Ongoing improvements in monitoring SOE debt, which is estimated at around 5 percent of GDP at end-2019 (excluding Petrocaribe-related obligations), should permit a broadening of the coverage of public debt, which is under consideration by the authorities. With Grenada's Petrocaribe arrangement essentially having ceased operations since the turn of 2017-18, we are in discussions with the authorities on the status of their obligations (which in the current DSA are included in external non-public debt).

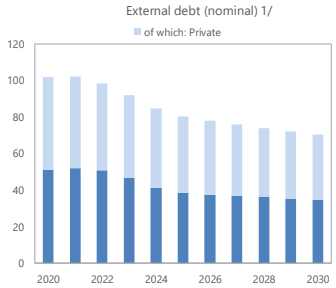
Table 1. Grenada: External Debt Sustainability Framework, Baseline Scenario 2017 – 2038

(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 9/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2038	Historical	Projections
External debt (nominal) 1/	**88.6**	**86.0**	**82.8**	**101.9**	**102.2**	**98.2**	**91.9**	**84.5**	**80.2**	**70.2**	**58.7**	**112.5**	**84.4**
of which: public and publicly guaranteed (PPG)	**47.5**	**45.1**	**41.6**	**51.1**	**51.8**	**50.6**	**46.6**	**41.1**	**38.4**	**34.8**	**31.0**	**59.6**	**41.8**
Change in external debt	-10.1	-2.5	-3.2	19.1	0.3	-4.0	-6.3	-7.3	-4.3	-1.8	-1.7		
Identified net debt-creating flows	**-4.7**	**-1.5**	**1.7**	**28.6**	**0.6**	**-2.3**	**-3.5**	**-5.1**	**-5.3**	**-3.8**	**-2.7**	**1.9**	**-0.6**
Non-interest current account deficit	**12.4**	**14.4**	**14.4**	**26.2**	**15.8**	**12.9**	**11.4**	**10.4**	**9.9**	**9.2**	**10.2**	**15.5**	**12.4**
Deficit in balance of goods and services	3.3	5.4	5.7	17.1	6.8	3.8	2.2	1.0	0.3	0.2	0.0	**10.4**	**3.0**
Exports	51.4	53.2	51.9	29.0	43.2	47.2	49.0	50.2	50.9	51.2	51.8		
Imports	54.7	58.6	57.6	46.1	50.0	51.0	51.2	51.1	51.2	51.4	51.8		
Net current transfers (negative = inflow)	0.3	0.8	0.4	0.5	0.6	0.8	0.9	0.9	1.0	1.2	1.2	**-0.8**	**0.9**
of which: official	0.4	0.5	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4		
Other current account flows (negative = net inflow)	8.8	8.2	8.3	8.6	8.4	8.3	8.3	8.5	8.7	7.8	9.0	**5.9**	**8.5**
Net FDI (negative = inflow)	**-13.5**	**-14.2**	**-10.6**	**-7.1**	**-10.6**	**-11.6**	**-12.6**	**-14.1**	**-14.1**	**-13.1**	**-12.2**	**-10.2**	**-12.3**
Endogenous debt dynamics 2/	**-3.6**	**-1.8**	**-2.0**	**9.5**	**-4.6**	**-3.6**	**-2.3**	**-1.3**	**-1.1**	**0.1**	**-0.7**		
Contribution from nominal interest rate	2.0	1.5	1.4	1.2	1.4	1.4	1.4	1.2	1.1	2.0	0.9		
Contribution from real GDP growth	-4.1	-3.5	-2.5	8.3	-5.9	-5.0	-3.6	-2.6	-2.2	-1.9	-1.6		
Contribution from price and exchange rate changes	-1.5	0.3	-1.0	...	...	...	...	...	...	...	...		
Residual 3/	**-5.4**	**-1.0**	**-4.9**	**-9.5**	**-0.3**	**-1.7**	**-2.9**	**-2.2**	**1.0**	**2.1**	**0.9**	**-4.9**	**-0.5**
of which: exceptional financing 4/	-5.5	0.0	0.0	-9.5	-2.4	-1.5	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**29.1**	**36.2**	**37.2**	**36.7**	**33.4**	**28.8**	**26.6**	**23.6**	**21.8**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**56.1**	**124.7**	**86.2**	**77.8**	**68.3**	**57.4**	**52.3**	**46.1**	**42.2**		
PPG debt service-to-exports ratio	**11.1**	**11.0**	**8.1**	**14.4**	**10.4**	**9.5**	**9.0**	**7.8**	**6.4**	**5.2**	**2.8**		
PPG debt service-to-revenue ratio	**24.8**	**24.4**	**17.8**	**19.0**	**19.8**	**19.3**	**18.6**	**16.6**	**13.8**	**11.3**	**6.1**		
Gross external financing need (Million of U.S. dollars)	52.7	71.9	98.1	259.2	117.1	75.5	44.5	3.3	-13.1	-22.6	-13.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.4	4.1	3.0	-9.2	6.3	5.3	3.9	3.0	2.7	2.7	2.7	**3.1**	**2.3**
GDP deflator in US dollar terms (change in percent)	1.5	-0.3	1.2	0.4	1.4	1.9	2.1	2.1	2.1	2.1	2.1	**1.6**	**1.8**
Effective interest rate (percent) 5/	2.2	1.8	1.7	1.3	1.4	1.5	1.5	1.4	1.3	2.9	1.5	**1.7**	**1.5**
Growth of exports of G&S (US dollar terms, in percent)	10.2	7.5	1.7	-49.0	60.7	17.0	10.1	7.7	6.4	5.0	4.8	**14.8**	**7.1**
Growth of imports of G&S (US dollar terms, in percent)	16.3	11.2	2.5	-27.0	17.0	9.3	6.5	5.0	5.0	5.0	5.0	**7.0**	**3.7**
Grant element of new public sector borrowing (in percent)	...	...	...	13.9	14.5	15.3	21.3	15.0	28.4	20.2	20.7	**...**	**18.0**
Government revenues (excluding grants, in percent of GDP)	23.0	23.9	23.7	21.9	22.6	23.2	23.6	23.6	23.6	23.6	23.7	**21.5**	**23.3**
Aid flows (in Million of US dollars) 6/	28.9	34.9	34.6	43.8	37.8	38.7	39.5	37.0	37.9	42.4	50.7		
Grant-equivalent financing (in percent of GDP) 7/	...	...	...	5.6	4.3	3.9	3.3	2.5	2.9	2.6	2.2	**...**	**3.3**
Grant-equivalent financing (in percent of external financing) 7/	...	...	...	35.8	39.6	45.0	68.8	121.1	73.6	64.1	62.2	**...**	**60.2**
Nominal GDP (Million of US dollars)	1,126	1,169	1,218	1,111	1,197	1,284	1,363	1,432	1,502	1,904	2,785		
Nominal dollar GDP growth	6.0	3.8	4.2	-8.8	7.8	7.3	6.1	5.1	4.9	4.9	4.9	**4.7**	**4.2**
Memorandum items:													
PV of external debt 8/	...	...	70.4	87.0	87.7	84.2	78.7	72.2	68.4	59.0	49.5		
In percent of exports	...	...	135.6	300.0	202.9	178.5	160.8	144.0	134.5	115.3	95.7		
Total external debt service-to-exports ratio	11.3	11.1	8.2	14.6	10.5	9.7	9.1	7.9	6.5	5.3	2.8		
PV of PPG external debt (in Million of US dollars)			354.5	401.6	445.9	471.1	455.3	412.0	399.7	449.7	608.4		
(PVt-PVt-1)/GDPt-1 (in percent)				3.9	4.0	2.1	-1.2	-3.2	-0.9	0.0	1.0		
Non-interest current account deficit that stabilizes debt ratio	22.5	16.9	17.6	7.1	15.5	17.0	17.7	17.7	14.3	11.0	11.9		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα(1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ For 2020 it includes all sources of exceptional financing under the RCF. For 2021 and beyond it includes Eximbank China lending to an SOE.

5/ Current-year interest payments divided by previous period debt stock.

6/ Defined as grants, concessional loans, and debt relief.

7/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

8/ Assumes that PV of private sector debt is equivalent to its face value.

9/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

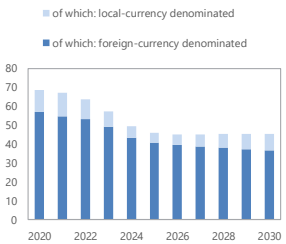
Table 2. Grenada: Public Sector Debt Sustainability Framework, Baseline Scenario 2017 – 2038

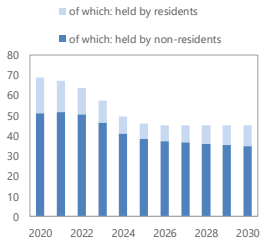
(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2038	Historical	Projections
Public sector debt 1/	**70.1**	**64.4**	**59.1**	**68.7**	**67.2**	**63.7**	**57.3**	**49.5**	**45.8**	**45.2**	**45.3**	**87.6**	**52.6**
of which: external debt	**47.5**	**45.1**	**41.6**	**51.1**	**51.8**	**50.6**	**46.6**	**41.1**	**38.4**	**34.8**	**31.0**	**59.6**	**41.8**
Change in public sector debt	-11.5	-5.8	-5.3	9.6	-1.4	-3.5	-6.4	-7.8	-3.7	0.0	0.0		
Identified debt-creating flows	**-7.7**	**-7.2**	**-7.5**	**9.8**	**-6.6**	**-7.7**	**-8.7**	**-8.2**	**-4.2**	**0.8**	**0.0**	**-3.0**	**-2.5**
Primary deficit	**-5.7**	**-6.6**	**-6.8**	**0.7**	**-4.6**	**-6.0**	**-6.7**	**-6.9**	**-3.0**	**0.6**	**0.7**	**-1.5**	**-2.2**
Revenue and grants	25.6	26.9	26.5	25.9	25.7	26.2	26.5	26.1	26.1	25.8	25.5	**24.4**	**26.0**
of which: grants	2.6	3.0	2.8	3.9	3.2	3.0	2.9	2.6	2.5	2.2	1.8		
Primary (noninterest) expenditure	19.9	20.3	19.7	26.6	21.1	20.3	19.8	19.3	23.1	26.5	26.1	**22.9**	**23.8**
Automatic debt dynamics	**-2.0**	**-0.6**	**-0.7**	**6.8**	**-3.6**	**-3.0**	**-2.0**	**-1.4**	**-1.2**	**0.2**	**-0.6**		
Contribution from interest rate/growth differential	-2.3	-1.8	-1.0	6.8	-3.6	-3.0	-2.0	-1.4	-1.2	0.2	-0.6		
of which: contribution from average real interest rate	1.2	0.9	0.8	0.8	0.4	0.4	0.4	0.3	0.2	1.4	0.6		
of which: contribution from real GDP growth	-3.5	-2.8	-1.9	6.0	-4.1	-3.4	-2.4	-1.6	-1.3	-1.2	-1.2		
Contribution from real exchange rate depreciation	0.3	1.3	0.3	…	…	…	…	…	…	…	…		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**2.3**	**1.7**	**1.2**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-0.2**	**0.5**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow 2/	0.0	0.0	0.0	2.3	1.7	1.2	0.0	0.0	0.0	0.0	0.0		
Residual 3/	**-3.8**	**1.4**	**2.3**	**-0.2**	**5.2**	**4.2**	**2.3**	**0.5**	**0.5**	**-0.8**	**0.0**	**-0.2**	**1.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 4/	**…**	**…**	**46.6**	**53.8**	**52.7**	**49.8**	**44.1**	**37.2**	**34.0**	**34.0**	**36.2**		
PV of public debt-to-revenue and grants ratio	**…**	**…**	**175.9**	**208.0**	**205.0**	**189.6**	**166.8**	**142.3**	**130.5**	**131.7**	**141.9**		
Debt service-to-revenue and grants ratio 5/	**59.3**	**56.4**	**34.5**	**19.2**	**30.4**	**24.3**	**20.9**	**15.9**	**10.6**	**15.3**	**20.7**		
Gross financing need 6/	9.5	8.6	2.3	7.9	4.9	1.6	-1.1	-2.7	-0.2	4.6	5.9		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.4	4.1	3.0	-9.2	6.3	5.3	3.9	3.0	2.7	2.7	2.7	**3.1**	**2.3**
Average nominal interest rate on external debt (in percent)	3.7	3.1	3.1	2.4	2.7	2.7	2.8	2.7	2.6	5.7	2.8	**2.9**	**2.9**
Average real interest rate on domestic debt (in percent)	1.5	2.9	1.5	2.8	1.5	1.5	1.3	0.9	0.3	1.9	3.2	**2.1**	**1.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	0.6	2.8	0.6	…	…	…	…	…	…	…	…	**0.2**	**…**
Inflation rate (GDP deflator, in percent)	1.5	-0.3	1.2	0.4	1.4	1.9	2.1	2.1	2.1	2.1	2.1	**1.6**	**1.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.1	6.2	-0.1	22.3	-15.6	1.2	1.5	0.4	22.9	2.5	2.6	**0.5**	**5.6**
Primary deficit that stabilizes the debt-to-GDP ratio 7/	5.8	-0.8	-1.5	-8.9	-3.2	-2.5	-0.2	0.9	0.7	0.6	0.6	**1.1**	**-0.9**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt . Definition of external debt is Residency-based.

2/ Reflects financing to SOEs from Exim Bank China and World Bank.

3/ Includes fluctuations of government bank deposits.

4/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

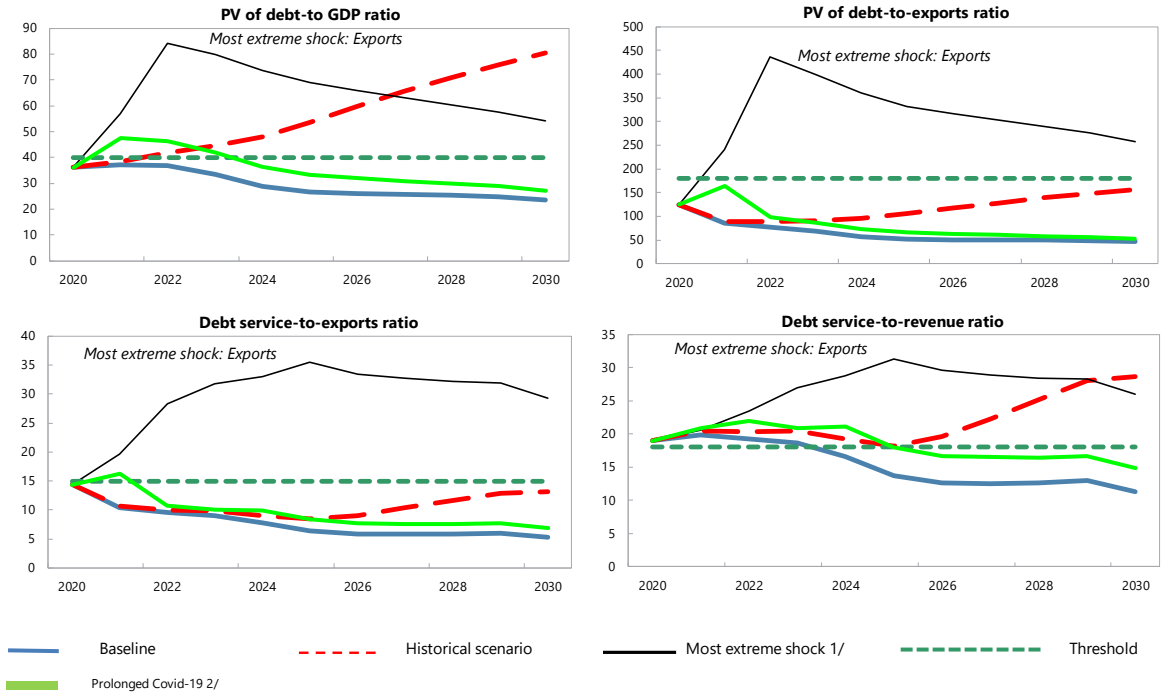
5/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

6/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

7/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Grenada: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2020-2030



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	No	Yes
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	3.5%	3.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	21	21
Avg. grace period	2	2

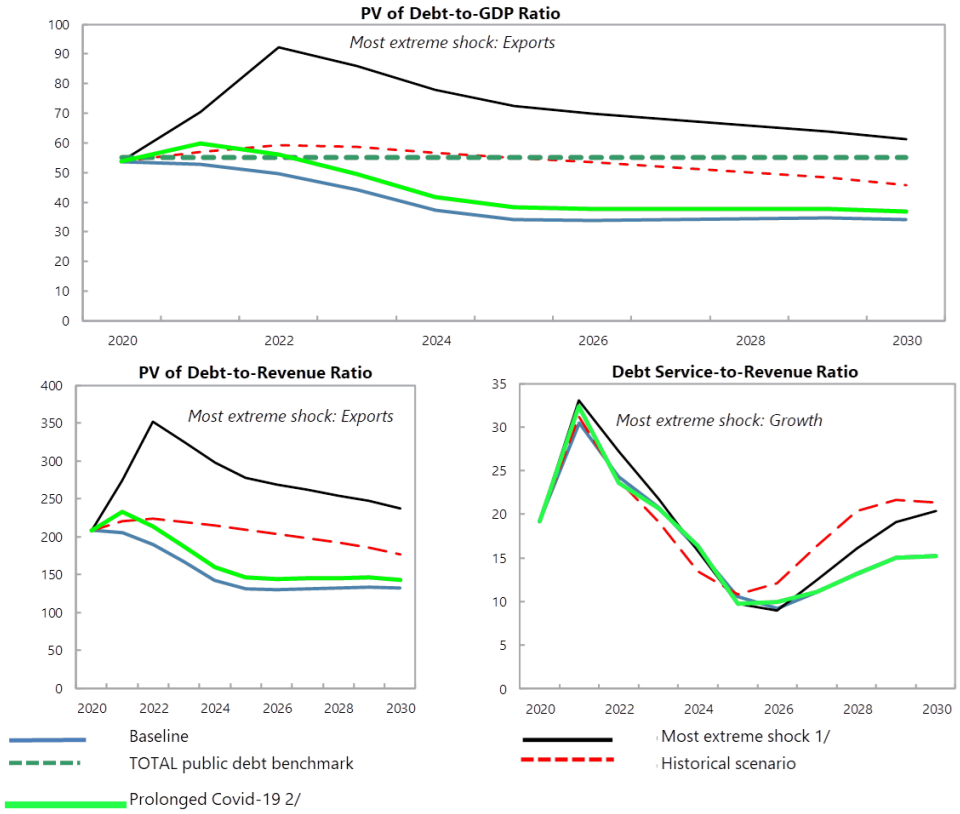
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ This scenario assumes that the currently projected effects of Covid-19 for 2020 remain similarly strong throghout 2021. Especifically, it assumes 5 percentage points lower real GDP growth in 2021 compared to the baseline and a one year delay in recovery afterwards, while exports, imports and FDI as a share of GDP remain at their 2020 level.

Figure 2. Grenada: Indicators of Public Debt under Alternative Scenarios, 2020-2030



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	152%	152%
Domestic medium and long-term	-47%	-47%
Domestic short-term	-5%	-5%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	3.5%	3.5%
Avg. maturity (incl. grace period)	21	21
Avg. grace period	2	2
Domestic MLT debt		
Avg. real interest rate on new borrowing	1.6%	1.6%
Avg. maturity (incl. grace period)	4	4
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	1.6%	1.6%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ This scenario assumes that the currently projected effects of Covid-19 for 2020 remain similarly strong throghout 2021. Specifically, it assumes 5 percentage points lower real GDP growth in 2021 compared to the baseline and a one year delay in recovery afterwards, while total public expenditure-to-GDP in 2021 remains equal to its 2020 value.

Table 3. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020-2030

(in percent, unless otherwise indicated)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	36	37	37	33	29	27	26	26	25	25	24
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	36	38	**42**	**45**	**48**	**54**	**60**	**65**	**71**	**76**	**80**
A2. Prolonged Covid-19	36	**48**	**46**	**42**	36	33	32	31	30	29	27
B. Bound Tests											
B1. Real GDP growth	36	**41**	**44**	40	34	32	31	30	30	29	28
B2. Primary balance	36	**47**	**56**	**50**	**44**	**40**	39	39	39	38	37
B3. Exports	36	**57**	**84**	**80**	**74**	**69**	**66**	**63**	**60**	**58**	**54**
B4. Other flows 3/	36	**43**	**48**	**45**	40	37	36	35	34	33	31
B5. Depreciation	36	**47**	**43**	39	33	30	30	29	29	29	27
B6. Combination of B1-B5	36	**62**	**67**	**63**	**56**	**52**	**50**	**49**	**47**	**45**	**43**
C. Tailored Tests											
C1. Combined contingent liabilities	36	**64**	**59**	**53**	**46**	**43**	**43**	**43**	**43**	**42**	**41**
C2. Natural disaster	36	**51**	**49**	**45**	39	37	37	37	37	37	36
C3. Commodity price	36	37	37	33	29	27	26	26	25	25	24
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	125	86	78	68	57	52	51	50	49	49	46
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	125	88	89	91	95	105	117	128	139	148	157
A2. Prolonged Covid-19	125	164	98	86	72	65	63	61	58	56	53
B. Bound Tests											
B1. Real GDP growth	125	86	78	68	57	52	51	50	49	49	46
B2. Primary balance	125	109	119	102	87	79	77	77	76	75	72
B3. Exports	125	**240**	**436**	**399**	**360**	**332**	**317**	**303**	**289**	**276**	**258**
B4. Other flows 3/	125	99	102	91	79	73	70	68	66	64	61
B5. Depreciation	125	86	72	63	52	47	46	46	45	45	43
B6. Combination of B1-B5	125	**204**	120	**204**	177	163	156	151	146	140	132
C. Tailored Tests											
C1. Combined contingent liabilities	125	149	125	108	93	84	84	84	84	83	80
C2. Natural disaster	125	120	104	92	79	72	73	73	72	72	70
C3. Commodity price	125	86	78	68	57	52	51	50	49	49	46
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	14	10	10	9	8	6	6	6	6	6	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	14	11	10	10	9	8	9	10	12	13	13
A2. Prolonged Covid-19	14	**16**	11	10	10	8	8	8	8	8	7
B. Bound Tests											
B1. Real GDP growth	14	10	10	9	8	6	6	6	6	6	5
B2. Primary balance	14	10	10	10	10	9	9	8	8	8	8
B3. Exports	14	**20**	**28**	**32**	**33**	**35**	**33**	**33**	**32**	**32**	**29**
B4. Other flows 3/	14	10	10	10	9	8	8	8	8	8	7
B5. Depreciation	14	10	10	9	8	6	5	5	5	6	5
B6. Combination of B1-B5	14	**17**	**21**	**20**	**21**	**18**	**17**	**17**	**17**	**17**	**15**
C. Tailored Tests											
C1. Combined contingent liabilities	14	10	12	11	9	8	7	7	7	7	6
C2. Natural disaster	14	11	11	10	9	7	7	7	7	7	6
C3. Commodity price	14	10	10	9	8	6	6	6	6	6	5
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	**19**	**20**	**19**	**19**	17	14	13	13	13	13	11
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**19**	**20**	**20**	**20**	**19**	**18**	**20**	**22**	**25**	**28**	**29**
A2. Prolonged Covid-19	**19**	**21**	**22**	**21**	**21**	18	17	16	16	17	15
B. Bound Tests	**19**	**21**	**22**	**21**	**21**	18	17	16	16	17	15
B1. Real GDP growth	**19**	**22**	**23**	**22**	**20**	16	15	15	15	15	13
B2. Primary balance	**19**	**20**	**21**	**22**	**21**	**20**	**18**	18	18	**18**	17
B3. Exports	**19**	**21**	**23**	**27**	**29**	**31**	**30**	**29**	**28**	**28**	**26**
B4. Other flows 3/	**19**	**20**	**20**	**20**	**20**	**18**	17	17	16	17	15
B5. Depreciation	**19**	**25**	**24**	**23**	**20**	16	15	15	15	15	13
B6. Combination of B1-B5	**19**	**22**	**26**	**26**	**28**	**25**	**23**	**23**	**23**	**23**	**21**
C. Tailored Tests											
C1. Combined contingent liabilities	**19**	**20**	**24**	**22**	**20**	16	15	15	15	16	14
C2. Natural disaster	**19**	**20**	**21**	**20**	**18**	15	14	14	14	15	13
C3. Commodity price	**19**	**20**	**19**	**19**	17	14	13	13	13	13	11
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold. The Prolonged Covid-19 scenario assumes that the currently projected effects of the pandemic for 2020 remain similarly strong throghout 2021.

2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

3/ Includes official and private transfers and FDI.

Table 4. Grenada: Sensitivity Analysis for Key Indicators of Public Debt, 2020-2030

(in percent, unless otherwise indicated)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	54	53	50	44	37	34	34	34	34	35	34
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	54	**57**	**59**	**59**	**57**	55	53	52	50	48	46
A2. Prolonged Covid-19	54	**60**	**56**	49	42	38	38	38	38	38	37
B. Bound Tests											
B1. Real GDP growth	54	**60**	**65**	**62**	**57**	**57**	**61**	**65**	**69**	**72**	**75**
B2. Primary balance	54	**59**	**63**	**57**	50	47	47	47	47	47	46
B3. Exports	54	**70**	**92**	**86**	**78**	**72**	**70**	**68**	**66**	**64**	**61**
B4. Other flows 3/	54	**58**	**61**	**55**	48	44	44	43	43	43	41
B5. Depreciation	54	**61**	**55**	48	40	36	34	32	31	29	27
B6. Combination of B1-B5	54	**58**	**58**	43	37	35	35	36	36	36	35
C. Tailored Tests											
C1. Combined contingent liabilities	54	**69**	**67**	**61**	54	51	51	51	51	51	51
C2. Natural disaster	54	**62**	**59**	54	47	44	44	45	45	46	46
C3. Commodity price	54	54	55	52	49	49	53	**58**	**62**	**66**	**70**
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	208	205	190	167	142	131	130	131	133	134	132
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	208	220	224	220	215	209	204	198	192	186	176
A2. Prolonged Covid-19	19	32	24	21	16	10	10	11	13	15	15
B. Bound Tests											
B1. Real GDP growth	208	229	241	228	215	215	229	245	260	276	287
B2. Primary balance	208	228	239	217	193	181	180	181	181	182	180
B3. Exports	208	274	351	325	298	277	268	261	254	247	237
B4. Other flows 3/	208	226	234	210	184	170	167	167	166	164	160
B5. Depreciation	208	238	213	184	155	137	130	124	119	114	105
B6. Combination of B1-B5	208	226	222	162	142	133	135	139	139	139	137
C. Tailored Tests											
C1. Combined contingent liabilities	208	269	253	231	208	197	195	196	198	199	197
C2. Natural disaster	208	240	225	203	180	170	170	172	175	177	176
C3. Commodity price	208	212	209	198	186	188	203	221	238	255	269
Debt Service-to-Revenue Ratio											
Baseline	19	30	24	21	16	11	9	11	13	15	15
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	19	31	24	19	13	11	12	16	20	22	21
A2. Prolonged Covid-19	19	32	24	21	16	10	10	11	13	15	15
B. Bound Tests											
B1. Real GDP growth	19	33	27	22	16	10	9	12	16	19	20
B2. Primary balance	19	30	20	15	15	11	12	16	17	18	19
B3. Exports	19	30	27	27	25	25	23	24	26	28	27
B4. Other flows 3/	19	30	25	22	19	14	13	15	17	18	19
B5. Depreciation	19	32	29	26	20	14	12	14	16	17	16
B6. Combination of B1-B5	19	30	25	17	13	8	7	11	13	15	15
C. Tailored Tests											
C1. Combined contingent liabilities	19	30	15	17	12	8	14	14	15	17	16
C2. Natural disaster	19	31	20	19	14	9	11	12	14	16	16
C3. Commodity price	19	30	23	19	14	8	8	10	13	16	17
Debt Service-to-GDP Ratio											
Baseline	5.0	7.8	6.4	5.5	4.2	2.8	2.4	2.9	3.4	3.9	3.9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	5	8	6	5	4	3	3	4	5	6	6
A2. Prolonged Covid-19											
B. Bound Tests											
B1. Real GDP growth	5	9	7	6	4	3	2	3	4	5	5
B2. Primary balance	5	8	5	4	4	3	3	4	4	5	5
B3. Exports	5	8	7	7	7	6	6	6	7	7	7
B4. Other flows 3/	5	8	7	6	5	4	3	4	4	5	5
B5. Depreciation	5	8	8	7	5	4	3	4	4	4	4
B6. Combination of B1-B5	5	8	7	4	3	2	2	3	3	4	4
C. Tailored Tests											
C1. Combined contingent liabilities	5	8	4	4	3	2	4	4	4	4	4
C2. Natural disaster	5	8	5	5	4	2	3	3	4	4	4
C3. Commodity price	5	8	6	5	4	2	2	3	3	4	4

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark. The Prolonged Covid-19 scenarios assume that the currently projected effects of the pandemic for 2020 remain similarly strong throghout 2021.

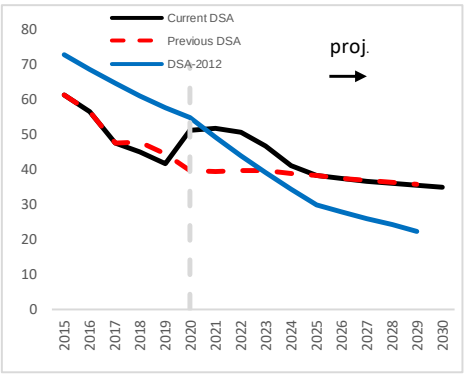
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.

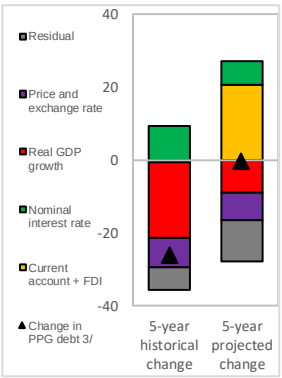
Figure 3. Grenada: Drivers of Debt Dynamics – Baseline Scenario

External debt

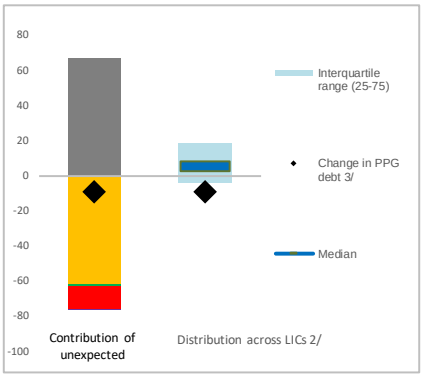
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

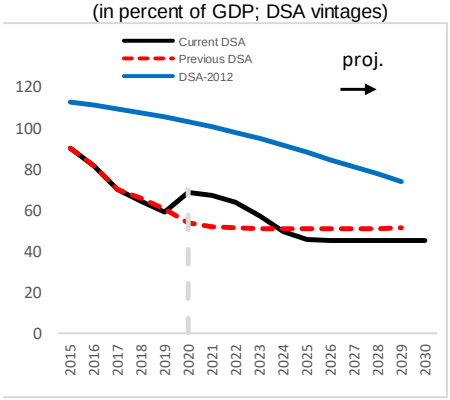


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

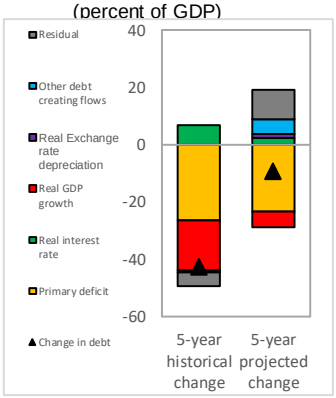


Public debt

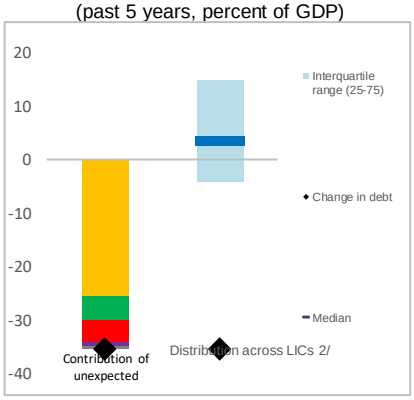
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



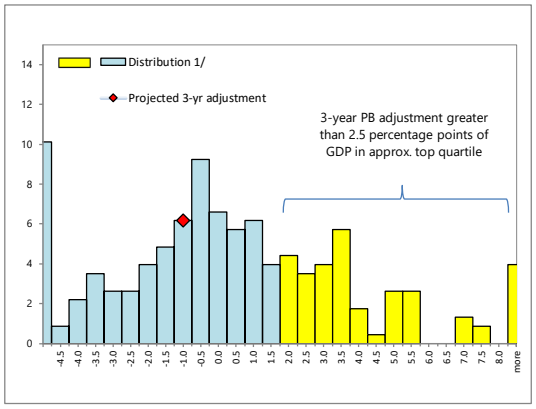
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

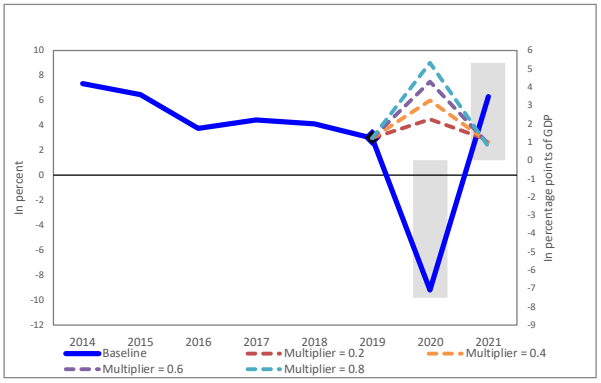
Figure 4. Grenada: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



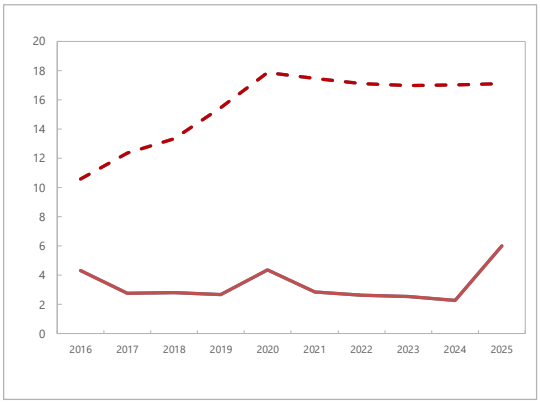
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



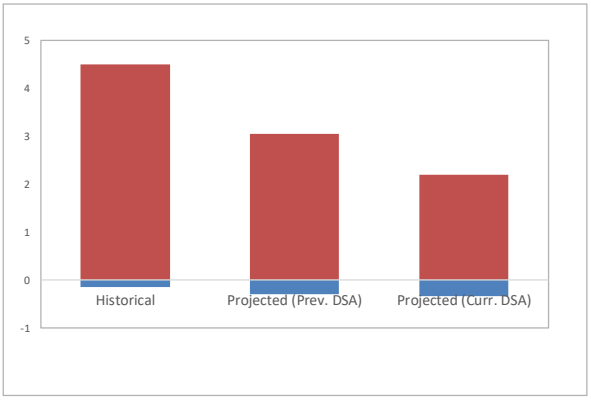
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Curr. DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Curr. DSA

Contribution to Real GDP growth
(percent, 5-year average)



Contribution of other factors
Contribution of government capital